

Most Frequent 2

Input file: **standard input**
Output file: **standard output**
Time limit: 1 second
Memory limit: 256 megabytes

Given an array of size n , print its most frequent element. If the max frequency value is x and more than one element has a frequency of x , then print -1.

Input

The first line contains a single number n ($1 \leq n \leq 1000$), the size of the array. Second line contains n numbers ($1 \leq n_i \leq 1000$) the array values.

Output

print a single integer. the most frequent element or -1.

Examples

standard input	standard output
5 1 2 3 4 5	-1
5 2 2 2 6 6	2

Note

see the first example:

- max frequency is 1.
- notice that all elements in the array has a frequency which is equal to the max frequency which violates our problem rule.

see the second example:

- frequency of 2 is 3 while frequency of 6 is 2
- since the max frequency is 3 and only one element has this frequency. the answer is 2